



IE3090

Network Programming

3rd Year, Semester I

Structured Reflection — IE3090 RemoteOps Assignment

Submitted to

Sri Lanka Institute of Information Technology

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Bachelor of Science Special Honors Degree in Information Technology

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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

Registration Number : IT24103381

Name : Jayarathne A W M R

AI tools used and when. I used Claude (Anthropic) for Part 1. I gave it the assignment brief and my registration number, and it generated the Agent, the Controller, the Makefile, a README and a test script, with the personalised values calculated from IT24103381: port 9410, SID tag 1833 and token OPS-3381. It also guided me step by step through setting up Linux, SSH, nano and scp, and through planning my screenshots and report. I used no other AI tool.

What the AI did well. It implemented the given protocol closely: line framing, exact byte counting for PUT and GET, a thread per Controller, a whitelist for EXEC, a UDP monitoring thread and timestamped logging. It reported running 33 automated checks, all passing, so I started from a working base rather than from nothing. When I got stuck, it re-explained things in simpler terms and in my own language.

Where did it go wrong or misled me. Its first suggestion for a Linux environment was WSL, but the installation failed on my network, so I used a CentOS Stream 10 machine over SSH instead. Some instructions led me to make mistakes: I ran a Windows command inside the CentOS terminal, and I typed a placeholder (IP_EKA) literally instead of a real IP address. My first scp attempt timed out, and it only worked later when I ran it from Windows PowerShell with the correct CentOS IP. Pasting the source files through nano also changed them: the checksums differed from the originals, with one extra byte in the Controller and two comment lines missing from the Makefile. I found this only by comparing file sizes and md5sum output, and I confirmed the differences were harmless.

What I changed, added or rejected. I did not submit the output untouched. I pasted each file into nano myself, built it using gcc 14.4.1, and ran the Agent and Controller. I verified the personalized values on my own machine using ss -tlnp for port 9410, the log file remoteops_IT24103381.log, and a listing of agentfiles/IT24103381/. All screenshots are from my own runs. I did not change the protocol logic, because I could not yet justify a change, and I rejected extra features I did not understand.

What surprised me. After my SSH session dropped, the Controller printed connect: Connection refused. The Agent had stopped with that session, so there was nothing listening. That showed me that the Controller and Agent are separate programs, and that the Agent has to keep running for anything to work. I was also surprised how exactness the protocol needs: every response must end with the right SID tag, and PUT and GET depend on exact byte count

What I learned. I saw how an Agent listens on a calculated port, requires AUTH first, and records every connection and command in its log, which I could read back in the log file. I also gained practical skills with SSH, scp and checksums. I cannot yet explain every function, for example how `read_line` handles partial lines, so before the Viva I will study the framing, PUT/GET and concurrency code until I can explain it without notes.

Next time. I will test each command as I build it rather than at the end, keep a design diary as I go, and run long-lived programs so that they survive a dropped SSH session.